



L2L Solutions Presents Bio-Pure

A Proven Enclosed Wastewater Treatment Facility
for Municipalities and Developments

High-Quality Effluent

Compact Footprint

Phased Capacity

Lower Conveyance Burden

Scalable

Municipal and Development Challenges



Aging Infrastructure

Replacement and rehabilitation backlogs compounding over decades

Tighter permit requirements straining outdated centralized plants

Failing legacy septic systems in older communities



Growth Pressure

High capital burden of extending sewers to new service areas

Uneven flow rates and seasonal shifts in developing zones

Economic development stalled by lack of immediate sanitation



Water Quality

Stricter nutrient (Nitrogen & Phosphorus) reduction mandates

Increasing municipal demand for non-potable water reuse

Enhanced protection for sensitive local ecosystems

EPA cites old infrastructure, rapid growth, and complex pollutants as the primary drivers straining municipal wastewater capacity.

Why Centralized Expansion Is Under Strain

Conventional centralized systems face critical bottlenecks

High Capital Constraints

Sewer extensions, lift stations, and deep trenching require massive upfront funding before any new homes are served.

The Conveyance Burden

Building miles of collection pipe before a single drop of water is treated adds enormous hidden lifecycle costs.

Mismatched Timelines

Long design, permitting, and construction phases mean municipal growth frequently outpaces actual plant capacity.



Eliminating the Conveyance Burden

Bio-Pure enables municipalities and developers to deploy decentralized satellite wastewater treatment facilities exactly where wastewater is generated — eliminating the need for costly sewer trunk line extensions across miles of undeveloped terrain.

By treating flows at the source, Bio-Pure dramatically reduces the conveyance burden, accelerates project deployment in active growth corridors, and allows infrastructure to be phased in lockstep with development absorption. The result is significantly lower capital expenditure, improved scalability, and a faster path from groundbreaking to certificate of occupancy.



Capital Heavy

Oversized infrastructure built for future 20-year projections demands major upfront financing.



Conveyance Network

Extensive piping networks suffer from infiltration/inflow and require ongoing costly maintenance.



Slow Deployment

Years-long timelines for environmental reviews, right-of-way acquisition, and construction.



Single Point of Failure

Centralizing treatment reduces resilience against extreme weather events and limits reuse options.

EPA and WEF cite lower upfront costs, modular scalability, and reduced footprint as primary advantages of decentralized wastewater systems.

Introducing Bio-Pure® — Proven Package Treatment



Bio-Pure is a factory-built wastewater treatment platform engineered for real-world municipal constraints — combining advanced biological treatment with unmatched modularity.

50+ Year History

First patented in 1972

NSF certified since 1973

Decades of continuous operational refinement

2,000+ Installations

Deployed successfully worldwide

Trusted across diverse operating environments

Proven longevity and field performance

Advanced ICEA Process

Intermittent-Cycle Extended Aeration

Completely enclosed, low-odor operation

Automated and remote operation capable

Deployment Flexibility

Above-ground or below-ground designs

Containerized and mobile form factors

Rapid installation on constrained sites

How the Technology Works

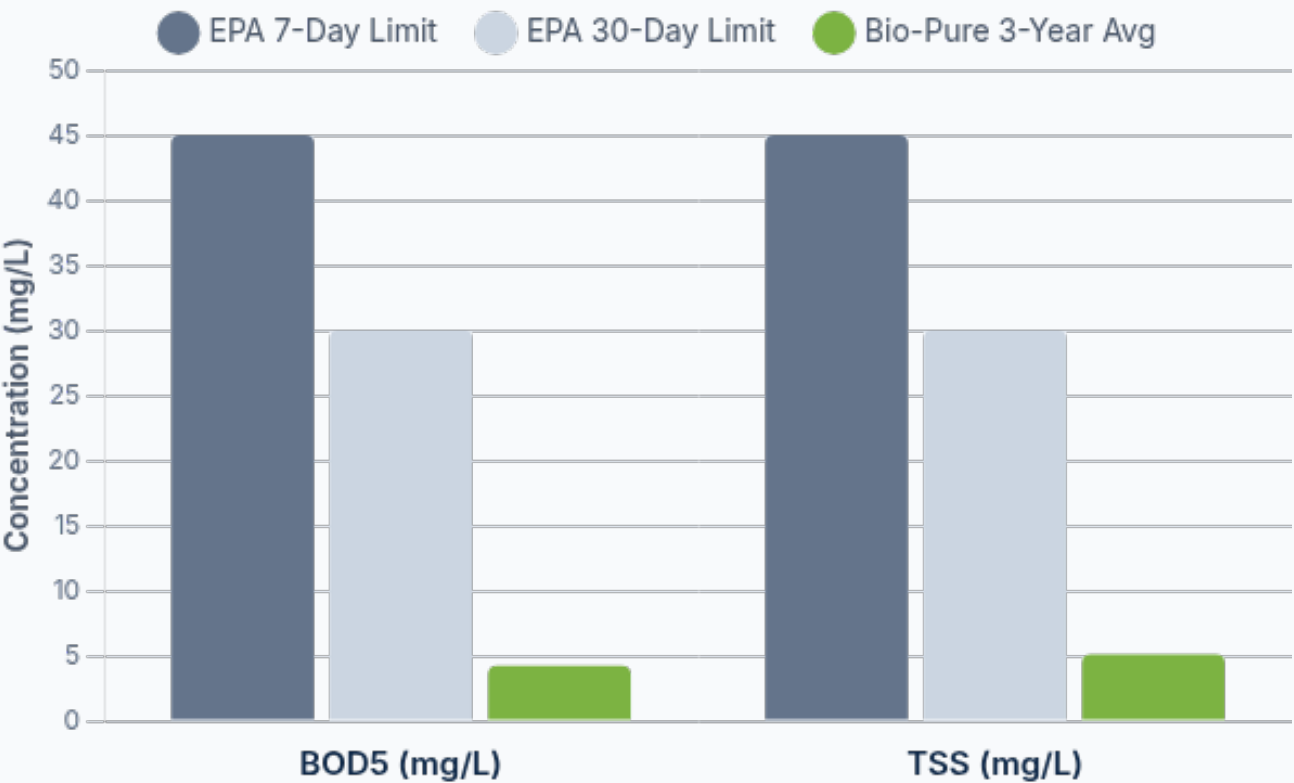


No chemical additives required • Anoxic denitrification capable • Configurable to any permit

Biology does the heavy lifting inside a controlled, enclosed process. Advanced polishing is added only when mandated by local reuse criteria.

Treatment Performance

BOD5 & TSS Effluent Comparison



40 CFR Part 133 secondary treatment limits mandate 30 mg/L (30-day avg) and 45 mg/L (7-day avg) for BOD5/TSS.

Coliform / Pathogen Control

<2.0 MPN/100mL

Meets or exceeds strict reuse standards, including CA Title 22 (2.2 MPN 7-day median).

Total Nitrogen Reduction

~1.7 mg/L

Typical effluent via advanced anoxic denitrification capability, drastically reducing nutrient loads to receiving waters.

Water Quality & Reuse Potential

Bio-Pure's high-quality effluent transforms wastewater into an asset, creating flexible discharge and reuse pathways.



Aquifer Support & Water Reuse

Treated influent can be further polished using integrated MBR (membrane bioreactor) technology to produce reclaimed water of a quality suitable for reintroduction into municipal water systems and aquifer recharge initiatives.

This positions Bio-Pure as a cornerstone of sustainable water management, enabling closed-loop reuse strategies that are particularly valuable for water-stressed municipalities, drought-prone regions, and high-growth developments seeking long-term water security.



Nonpotable Reuse

Ideal for landscape irrigation, golf courses, parks, and water features.

Reclaimed water offsets potable demand and supports sustainable community development while providing reliable green space hydration year-round.



Surface Discharge

High-quality effluent protects receiving waters, ensuring environmental compliance and supporting local aquatic ecosystems.

Safe discharge is achieved naturally without the need for toxic chemical residuals.

California Title 22 disinfected tertiary recycled water requires a 7-day median coliform of ≤ 2.2 MPN/100mL and strict turbidity targets. Bio-Pure's optional MBR polishing ensures compliance with the most stringent municipal reuse standards.

Environmental Benefits



Reduced Nutrient Loading

Anoxic denitrification lowers nitrogen and phosphorus levels, preventing eutrophication and protecting downstream water quality.



Enclosed, Odor-Free

Completely enclosed process minimizes community nuisance and visual impact, allowing integration closer to residential growth nodes.



Ecosystem Protection

High-quality effluent protects sensitive local waterways, ensuring municipal regulatory compliance and safeguarding natural habitats.

Water Sustainability

Transforms municipal wastewater into a reusable asset for landscape irrigation or local aquifer recharge, supporting circular water initiatives.



Energy & Conveyance

Avoids the massive energy burden and lifecycle costs of pumping wastewater across miles of long sewer runs by treating closer to the source.



Distributed Resilience

Adding decentralized capacity makes the municipal network less vulnerable to single-point-of-failure events and extreme weather.

Wastewater infrastructure investment aligns with municipal sustainability targets, ESG mandates, and regional watershed protection goals.

Biosolids & Operations Advantage

The Bio-Pure Advantage

95%

Solids Volume
Reduction

Extensive internal recycle minimizes generated waste sludge

Annual desludging drastically reduces maintenance hours

Flexible handling: drying beds, geotextile bags, or standard hauling

Enclosed process eliminates open-air odor complaints

Conventional Pain Points

- ✗ **High volume generation** requires constant operator attention
- ✗ **Frequent hauling costs** and rising landfill tipping fees
- ✗ **Expensive dewatering equipment** and higher O&M
- ✗ **Community nuisance** from open-air handling



Every gallon of sludge not hauled is a direct win for the city's operating budget, staff bandwidth, and environmental footprint.

Compact Footprint & Siting Flexibility

Bio-Pure systems minimize visual and spatial impact, allowing municipalities to build capacity exactly where it's needed.



Above-Ground

Most cost-effective and easily accessible configuration for operations and maintenance. Ideal for secure municipal properties, dedicated utility zones, and remote satellite districts.



Below-Ground

Provides maximum aesthetic integration and space-saving footprint. Enclosed operation ensures odor control, making it perfect for constrained residential areas, parks, or mixed-use developments.



Containerized / Mobile

Housed securely in standard shipping containers. Enables rapid, modular deployment for temporary work camps, emergency relief scenarios, or phased infrastructure scale-ups.



Fit Constrained Sites

Compact footprint unlocks unusable or non-perking land, reducing the need for massive, expensive municipal land acquisitions.



Shallower Collection

Distributed placement allows alternative sewer lines to be placed at shallower depths, avoiding costly deep trenching and complex lift stations.

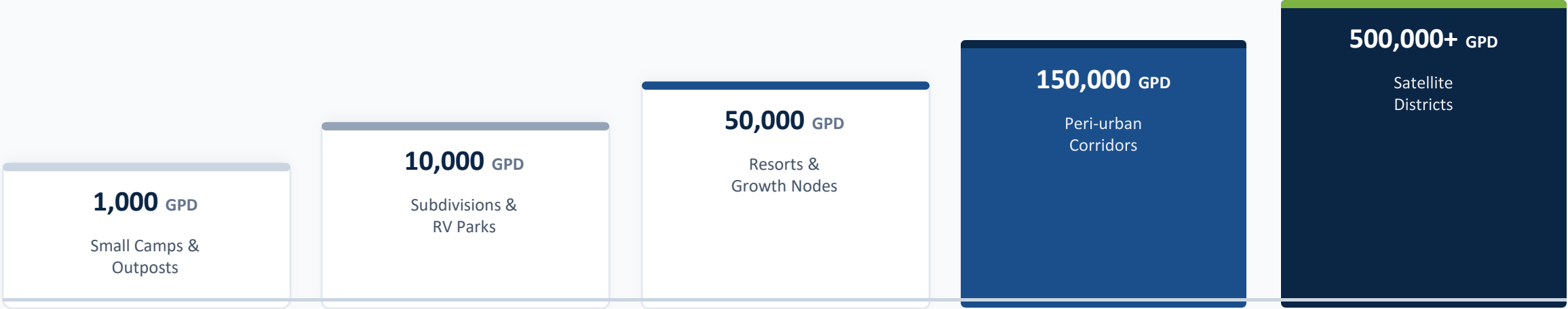


Capacity Near Demand

Treat wastewater exactly where it is generated (growth nodes, campuses, or resorts), slashing extensive and wasteful conveyance burdens.

Scalability & Phased Municipal Growth

Phase investment with actual development. Add treatment capacity only when needed to avoid massive up-front financing.



Phase Investment

Match capital outlay exactly with actual development. Avoid bonding for massively oversized central plants on day one before tax revenues arrive.



Operational Flexibility

Handles even or uneven flows flawlessly. The ICEA process maintains identical biological process efficiency from 10% to 100% of design capacity.

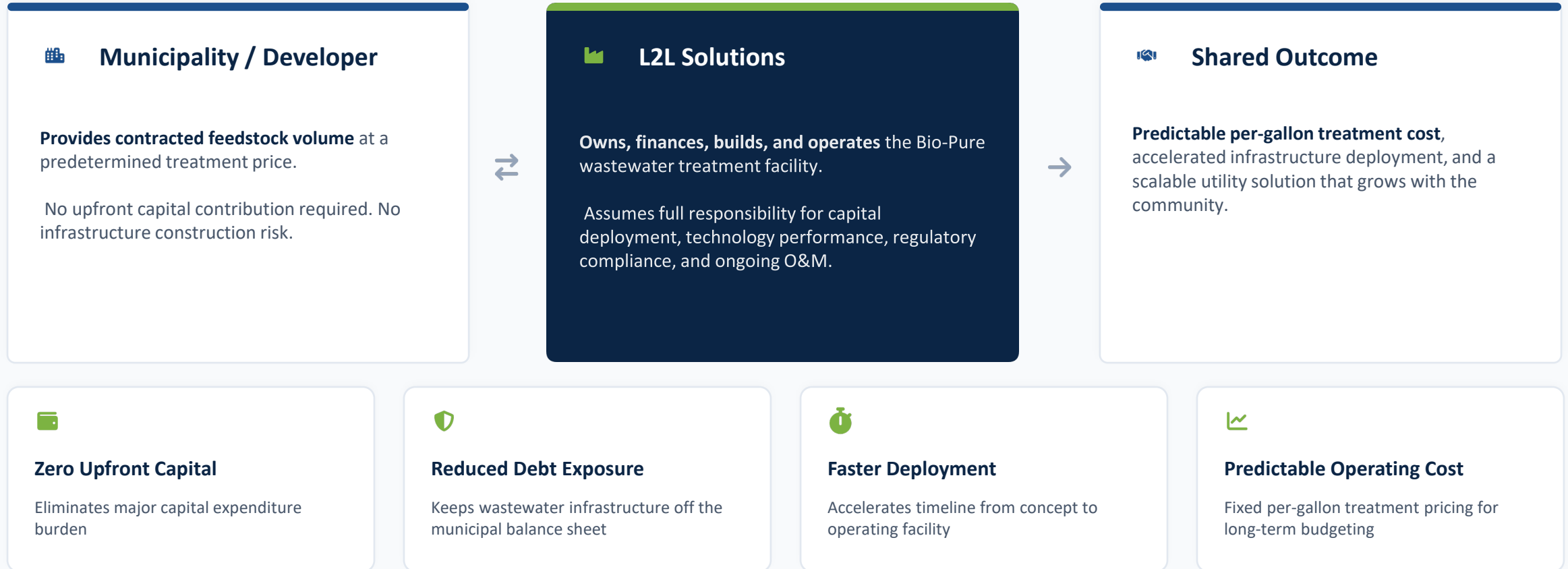


Remote Capability

Systems under 50,000 gpd can be operated entirely remotely via advanced CPU controls, drastically slashing required on-site operator hours and staffing burdens.

Ownership & Delivery Model — Utility-as-a-Service

A Public-Private Partnership (PPP) framework that eliminates municipal capital burden



"A turnkey utility-as-a-service model purpose-built for municipalities and developers seeking institutional-grade wastewater capacity without institutional-grade capital risk."

How Bio-Pure Compares

Evaluation Criteria	Lagoon	Activated Sludge	Bio-Pure®	Full MBR
Footprint	Very large	Moderate to large	Compact / Modular	Very compact
Sewer Extension Need	High	High (centralized)	Lower (decentralized)	High (centralized)
Effluent Quality	Variable	Good to very good	High; optional polishing	Excellent
Sludge Burden	Moderate	Moderate	Positioned as low	Lower solids; higher O&M
Reuse Readiness	Limited	With polishing	Strong (permit-dependent)	Strong
Expansion Style	Land-intensive	Capital-intensive	Phased / Modular	High-tech, higher cost
Best Fit	Land-rich, small towns	Dense service areas	Growth nodes, constrained sites	Tight permits, premium reuse

A practical alternative between land-intensive lagoons and capital-intensive advanced plants.

Use Cases & Proof of Performance



RV Park & Golf Course

📍 California • Operating since 1993



Shopping Center

📍 California • Operating since 2003



Coastal RV Park

📍 Oregon • Operating since 1986



Public Campground

📍 Oregon • Operating since 1986

Ideal Municipal Fits

Peri-Urban Growth Corridors

Satellite Utility Districts

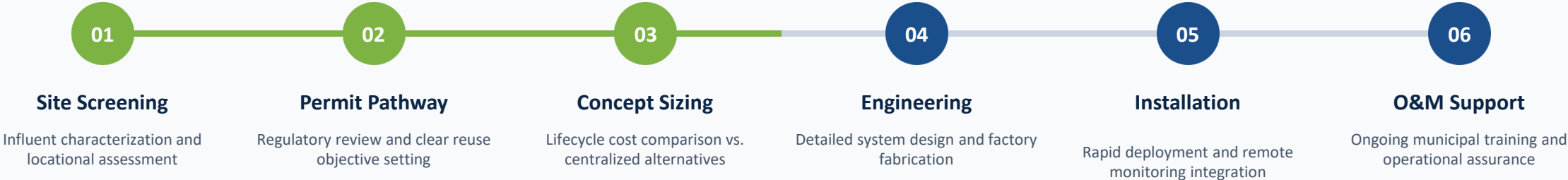
Public Campuses & Institutions

Parks & Resort Developments

Constrained Redevelopment Sites

"Proven reliability across decentralized use cases where conventional septic fails and large sewer extensions are cost-prohibitive."

Implementation Roadmap & Call to Action



Recommended Next Steps

Identify Target Projects

1

Locate one current high-cost sewer-extension or a pressing capacity bottleneck project to evaluate immediate Bio-Pure feasibility.

Run the Comparison

2

Work with our engineers on a side-by-side conceptual comparison against centralized upgrades or sprawling pipeline expansions.

Assess Reuse Value

3

Review local discharge regulations and aquifer targets to determine if optional MBR polishing is warranted for your community.

L2L Solutions — Bio-Pure: **Decentralized infrastructure that scales with your community.**